HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Python** data pipelines for large-scale audio dataset processing, enabling model training for a team of 5 AI scientists.
- Engineered **Bash** scripts and **Linux** shell commands to automate data ingestion and transformation workflows across distributed file systems, reducing manual effort by an estimated 40%.
- Collaborated with **AI scientists** to define and implement data quality checks and validation frameworks for audio features, improving data integrity for downstream model development.
- Designed and deployed **data processing systems** on **AWS** infrastructure, including EC2 and S3, to handle terabytes of audio data for model training.
- Containerized data pipeline components using **Docker**, ensuring consistent execution environments and simplifying deployment to cloud infrastructure.
- Implemented **REST APIs** using **FastAPI** to serve processed audio data features for consumption by AI model training services, achieving p95 latencies under 200ms.
- Managed **CI/CD pipelines** using **Jenkins** for automated testing and deployment of data processing jobs, improving deployment frequency by 3x.

SKILLS

Languages – Python, Bash, SQL, Java

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Linux

Data Engineering & Processing – Data Pipelines, ETL, Large-Scale Data Processing, Data Quality, Data Validation, Feature Engineering

AI & ML – AI Model Training, Audio Data Processing, Machine Learning Fundamentals

Development Tools – Git, Jenkins, Jira, Confluence, REST APIs, FastAPI, Spring Boot

PROJECTS

Scalable Audio Data Processing Pipeline

- Built a **Python**-based data pipeline to ingest, clean, and transform large volumes of raw audio files for AI model training.
- Implemented parallel processing on **AWS EC2** instances to accelerate data transformation tasks, reducing processing time for a 1TB dataset by 60%.
- Developed a **data validation** module within the pipeline to ensure feature consistency and quality, catching approximately 15% of data anomalies before model ingestion.

AI Model Training Data Preparation Framework

- Designed and implemented a modular framework using **Python** and **Pandas** for preparing diverse audio datasets for machine learning model training.
- Integrated feature extraction scripts for audio characteristics, creating a feature set that enhanced model performance by an average of 8% across several test models.
- Wrote comprehensive documentation and runbooks for the data preparation framework, enabling faster onboarding for new AI scientists joining the team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)